Average Causal Effect in Observational Studies III

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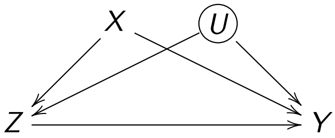
Spring 2026

Identification assumptions in observational studies

- Identification assumptions:
  - ignorability: $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid \mathbf{X}_i$
  - overlap: $0 < \text{pr}(Z_i = 1 \mid \mathbf{X}_i) < 1$ for all $\mathbf{X}_i$

- Possible existence of observed and unobserved confounders

$$\mathbb{E}\{Y_i(z) \mid Z_i = 1, \mathbf{X}_i\} \neq \mathbb{E}\{Y_i(z) \mid Z_i = 0, \mathbf{X}_i\}$$



- Latent ignorability: $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid (\mathbf{X}_i, U_i)$
- Untestable from the observed data

Causal diagram

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- Useful for visualizing causal relationship

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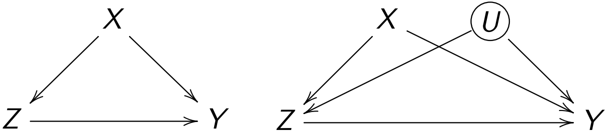

$$X \sim F_X(x), \quad Z = f_Z(X, \epsilon_Z), \quad Y(z) = f_Y(X, z, \epsilon_Y(z))$$

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\qquad
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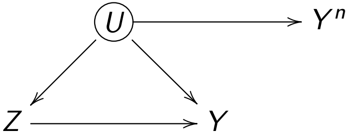
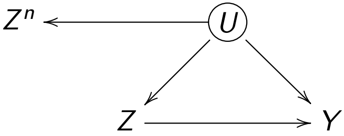
Negative outcome



- Negative outcome Y_i^n: similar to Y_i in terms of confounding
  - if $Z_i \perp\!\!\!\perp \{Y_i(1), Y_i(0)\} \mid \mathbf{X}_i$, then $Z_i \perp\!\!\!\perp \{Y_i^n(1), Y_i^n(0)\} \mid \mathbf{X}_i$
  - $\mathbb{E}\{Y_i^n(1) - Y_i^n(0)\} =$ known value
- Assessing ignorability $\rightsquigarrow$ estimate ACE on the negative outcome
- Examples:
  - effect of smoking on car accident, violence, suicide (Cornfield et al., 1959)
  - lagged outcome (Imbens and Rubin, 2015)
  - Lipsitch et al., (2010), "Negative Controls: A Tool for Detecting Confounding and Bias in Observational Studies"
  - the effect of known effects (Rosenbaum, 1989)

Negative treatment



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  - $\mathbb{E}\{Y_i(Z_i^n = 1) - Y_i(Z_i^n = 0)\} =$ known value

- Assessing ignorability $\rightsquigarrow$ estimate ACE of the negative treatment

- Examples: Sanderson et al. (2018) "Negative control exposure studies in the presence of measurement error: implications for attempted effect estimate calibration"

Negative treatment

Exposure	Negative control exposure	Outcome(s)
Maternal smoking	Paternal smoking	Offspring outcomes: Inattention/hyperactivity[15,20] Obesity/adiposity[16,22–24] Blood pressure[17] Gestational diabetes[21] ADHD symptoms[19] Cognitive development[18] Offspring psychotic symptoms[46]
Maternal psychosocial stress	Paternal psychosocial stress	Offspring vascular function[54]
Maternal smoking during pregnancy	Maternal smoking after pregnancy	Offspring respiratory outcomes[39] Offspring psychotic symptoms[46]
Maternal alcohol consumption during pregnancy	Maternal alcohol consumption before pregnancy	Offspring ADHD symptoms[40]
Maternal BMI/obesity	Paternal BMI	Offspring BMI/adiposity[26–33] Offspring cognitive and psychomotor development[55]
Length of pre-birth inter-pregnancy interval	Length of post-birth inter-pregnancy interval	Risk of schizophrenia in the offspring[56]
Folic acid supplements in pregnancy	Other supplements in pregnancy	Autism spectrum disorders[37] Language development delays[38]
Prescription for trimethoprim 1–3 months before pregnancy	Prescription for trimethoprim 13–15 months before pregnancy	Offspring congenital malformation[57]
Air pollutant exposure during pregnancy	Air pollutant exposure before and after pregnancy	Offspring autism spectrum disorder[41]
Exposure to childhood infections	Hospital attendance for broken bones	Multiple sclerosis later in life[58]
Adherence to prescribed statins and beta blockers	Adherence to other prescribed medication	Long-term mortality after acute myocardial infarction[59]
Vaccination during flu season	Vaccination outside flu season	Mortality and hospitalization from flu[60]
Swimmers' exposure to bacteria in water	Non-swimmers	Gastrointestinal illnesses after an increase in bacteria levels in water[61]

Problem of over-adjustment

- Rosenbaum (2002): "there is no reason to avoid adjustment for a variable describing subjects before treatment"
- Rubin (2007): typically, the more conditional an assumption, the more acceptable it is.
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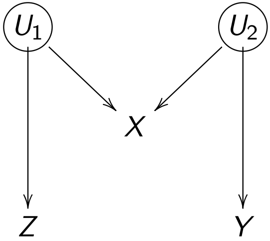
- Pearl disagrees

M-bias



- Linear model:

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X &= aU_1 + bU_2 + \epsilon_X \\
Z &= cU_1 + \epsilon_Z \\
Y(z) &= dU_2 + \epsilon_Y \\
(U_1, U_2, \epsilon_X, \epsilon_Z, \epsilon_Y) &\sim i.i.d.\ N(0,1)
\end{aligned}
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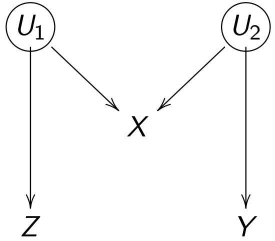

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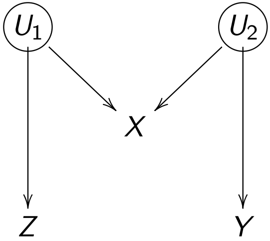

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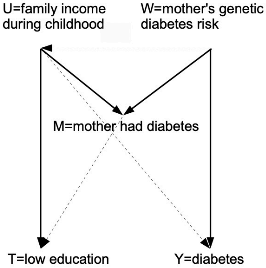
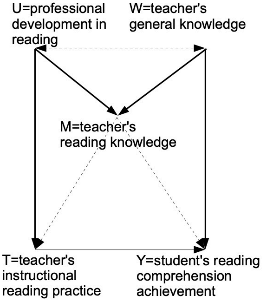
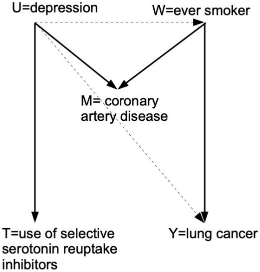
M-structures with deviations



(a) Glymour (2006)



(b) Kelcey and Carlisle (2011)



(c) Liu et al. (2012)

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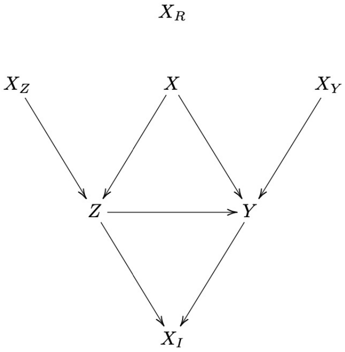
What covariates should be adjusted?



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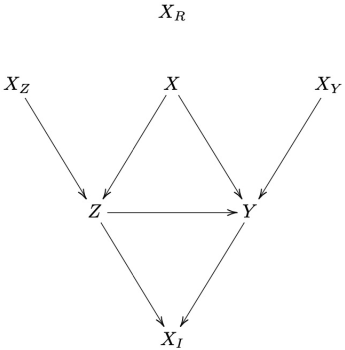

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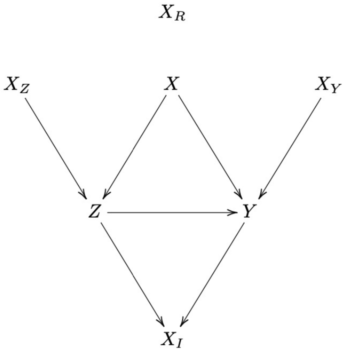

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- Sensitivity analysis: how results would change under certain types of latent confounding
  - nonparametric sensitivity analysis, e.g., Cornfield condition
  - parametric sensitivity anlaysis

Partial identification

- Identification $\rightsquigarrow$ point estimation
- Partial identification
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- Cochran (1953) uses the idea of worse-case bounds in surveys with missing data
- Manski (1990) applies the idea to causal inference

Bounds on ACE

- Outcome bounded in $[l, u]$
- Lower and upper bound on ACE

$$\mathbb{E}(Y_i \mid Z_i = 1)\Pr(Z_i = 1) + l\Pr(Z_i = 0) - u\Pr(Z_i = 1) - \mathbb{E}(Y_i \mid Z_i = 0)\Pr(Z_i = 0)$$
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- Bounds are often too wide to be informative

- Stronger assumptions $\rightsquigarrow$ tighter bounds
  - other possible assumptions: $\mathrm{cor}(Y(1), Y(0)) \geq 0$, $Z = \mathbf{1}(Y(1) - Y(0) \geq 0)$
  - statistical inference: confidence interval on the bounds or the true parameter (Imbens and Manski, 2004)

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  - nonparametric sensitivity analysis for binary outcome $\rightsquigarrow$ Cornfield condition
  - parametric sensitivity analysis

Smoking and lung cancer

- Z: smoking
- Y: lung cancer
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- Fisher (1957) criticizes their result because it is possible that a hidden gene (U) simultaneously causes cigarette smoking and lung cancer although the true causal effect of cigarette smoking on lung cancer is absent

Sensitivity analysis (Cornfield et al. 1959. *J. Natl. Cancer Inst.*)

- Question: How important does a confounder have to be to explain away the observed association? $\rightsquigarrow$ robustness of findings

The magnitude of the excess lung-cancer risk among cigarette smokers is so great that the results can not be interpreted as arising from an indirect association of cigarette smoking with some other agent or characteristic, since this hypothetical agent would have to be at least as strongly associated with lung cancer as cigarette use, no such agent has been found or suggested.

Cornfield Condition

- Causal and observed relative risks:

$$RR_{ZY} = \frac{\Pr\{Y_i(1) = 1\}}{\Pr\{Y_i(0) = 1\}}, \quad RR_{ZY}^{obs} = \frac{\Pr(Y_i = 1 \mid Z_i = 1)}{\Pr(Y_i = 1 \mid Z_i = 0)}$$

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- How large do RR_{UY} and RR_{ZU} have to be in order for $RR_{ZY}^{obs} > 1$ and $RR_{ZY} = 1$ hold at the same time?

Cornfield Conditions

Theorem

Suppose that U is binary and $Z \perp\!\!\!\perp Y \mid U$. Assume

$$RR^{obs}_{ZY} > 1, \quad RR_{ZU} > 1, \quad RR_{UY} > 1.$$

We have

$$RR^{obs}_{ZY} \;\leq\; \frac{RR_{ZU} \times RR_{UY}}{RR_{ZU} + RR_{UY} - 1}$$

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- Define $h(w_1, w_2) = w_1 w_2 / (w_1 + w_2 - 1)$ for $w_1 > 1$ and w_2
  - $h(w_1, w_2) \le \min(w_1, w_2)$
  - $h(w_1, w_2) \le w^2/(2w - 1)$, where $w = \max(w_1, w_2)$

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  - $h(w_1, w_2) \le w^2/(2w - 1)$, where $w = \max(w_1, w_2)$
- Cornfield condition: $\min(RR_{ZU}, RR_{UY}) \ge RR_{ZY}^{\text{obs}}$

E-value

- $\max(RR_{ZU}, RR_{UY}) \geq RR_{ZY}^{\text{obs}} + \sqrt{RR_{ZY}^{\text{obs}}(RR_{ZY}^{\text{obs}} - 1)}$

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- E-value (VanderWeele and Ding, 2017): $RR_{ZY}^{\text{obs}} + \sqrt{RR_{ZY}^{\text{obs}}(RR_{ZY}^{\text{obs}} - 1)}$
  - the maximum of the confounding measures RR_{UY} and RR_{ZU} need to be at least as large as the E-value to explain away the observed relative risk
  - *p*-values: sampling uncertainty
  - E-value: confounding bias

Smoking and Lung Cancer

	Lung cancer	No lung cancer
Smoker	397	78577
Non-smoker	51	108778

- $RR_{ZY}^{obs} = 10.73$ with 95% confidence interval $[8.02, 14.36]$
  (Hammond and Horn. 1958. *JAMA*)

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$$\min(RR_{ZU}, RR_{UY}) \geq 10.73$$

- Generalized Cornfield condition:

$$\max(RR_{ZU}, RR_{UY}) \;\geq\; RR^{obs}_{ZY} + \sqrt{RR^{obs}_{ZY}(RR^{obs}_{ZY} - 1)} \;=\; 20.95$$

- For the lower confidence interval, this number equals 15.52

Extensions

- Non-zero true causal effect

$$RR_{ZY} \;\leq\; RR_{ZY}^{obs} \times \frac{RR_{ZU} + RR_{UY} - 1}{RR_{ZU} \times RR_{UY}}$$

  - modify the definition
    $RR_{UY} = \max_z \Pr(Y = 1 \mid Z = z, U) / \Pr(Y = 1 \mid Z = z, U = 0)$
  - upper bound on RR_{ZY} with two sensitivity parameters

Extensions

- Non-zero true causal effect

$$RR_{ZY} \ \leq \ RR_{ZY}^{obs} \times \frac{RR_{ZU} + RR_{UY} - 1}{RR_{ZU} \times RR_{UY}}$$

  - modify the definition
    $RR_{UY} = \max_z \Pr(Y = 1 \mid Z = z, U) / \Pr(Y = 1 \mid Z = z, U = 0)$
  - upper bound on RR_{ZY} with two sensitivity parameters

- Discrete U: Ding and VanderWeele (2016)

Parametric sensitivity analysis

- Include U in regression models (Rosenbaum and Rubin, 1983)

$$\begin{aligned}\operatorname{logit}\Pr\{Y_i(z) \mid \mathbf{X}_i, U_i\} &= \beta_0 + \beta_Z z + \beta_X \mathbf{X}_i + \beta_U U_i \\ \operatorname{logit}\Pr(Z_i = 1 \mid \mathbf{X}_i, U_i) &= \alpha_0 + \alpha_X X_i + \alpha_U U_i\end{aligned}$$

  - binary U with sensitivity parameters $\Pr(U = 1 \mid \mathbf{X}_i)$
  - can use other models and sensitivity parameters
  - becomes very arbitrary sometimes

Sensitivity analysis for ACE

- Sensitivity parameters $\epsilon_1(X)$ and $\epsilon_0(X)$

$$
\begin{aligned}
\frac{\mathbb{E}\{Y(1) \mid Z = 1, X\}}{\mathbb{E}\{Y(1) \mid Z = 0, X\}} &= \epsilon_1(X) \\
\frac{\mathbb{E}\{Y(0) \mid Z = 1, X\}}{\mathbb{E}\{Y(0) \mid Z = 0, X\}} &= \epsilon_0(X)
\end{aligned}
$$

- $\epsilon_1(X) = \epsilon_0(X) = 1$ implies ignorability
- observed data provide no information on the sensitivity parameters

Sensitivity analysis for ACE

- Outcome regression formula

$$
\begin{aligned}
\mathbb{E}\{Y(1) \mid Z=1\} &= \mathbb{E}\{\mu_1(X)/\epsilon_1(X) \mid Z=1\} \\
\mathbb{E}\{Y(0) \mid Z=0\} &= \mathbb{E}\{\mu_0(X)\epsilon_0(X) \mid Z=0\}
\end{aligned}
$$

- Inverse probability weighting formula

$$
\begin{aligned}
\mathbb{E}\{Y(1)\} &= \mathbb{E}\left\{w_1(X)\frac{ZY}{e(X)}\right\} \\
\mathbb{E}\{Y(0)\} &= \mathbb{E}\left\{w_0(X)\frac{(1-Z)Y}{1-e(X)}\right\}
\end{aligned}
$$

  - $w_1(X) = e(X) + \{1-e(X)\}/\epsilon_1(X)$ and $w_0(X) = e(X)\epsilon_0(X) + \{1-e(X)\}$

Summary

- Assessing ignorability: negative outcome, negative treatment
- When ignorability fails
    - partial identification
    - sensitivity analysis
- Other strategies
    - instrumental variable method
    - twin study, DID, etc.

Suggested readings

- Sensitivity analysis
  - parametric: Rosenbaum and Rubin, "Assessing Sensitivity to an Unobserved Binary Covariate in an Observational Study with Binary Outcome"
  - non-parametric: Ding's book, Ding and VanderWeele, "Sensitivity Analysis Without Assumptions"

- Bradford Hill criteria: nine criteria to provide epidemiologic evidence of a causal relationship
  - strength $\rightsquigarrow$ cornfield condition, E-value
  - consistency $\rightsquigarrow$ meta-analysis, invariant prediction (Peters, Buhlmann and Meinshausen, 2016)
  - Specificity $\rightsquigarrow$ specificity score